

3.1 Newton's Laws of Motion

Question Paper

Course	CIEA Level Physics
Section	3. Dynamics
Topic	3.1 Newton's Laws of Motion
Difficulty	Medium

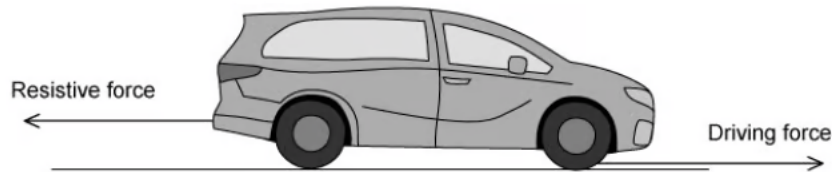
Time allowed: 20

Score: /10

Percentage: /100

Question 1

A car has a horizontal driving force of 2.0 kN acting on it and a resistive force a quarter of this size. It has a forward horizontal acceleration of 2.0 m s^{-2}



What is the mass of the car?

- A. 500 kg
- B. 750 kg
- C. 1250 kg
- D. 2000 kg

[1 mark]

Question 2

All external forces on a body cancel out.

Which statement **must** be correct?

- A. the body does not move
- B. the total energy (kinetic and potential) of the body remains unchanged
- C. the speed of the body remains unchanged
- D. the momentum of the body remains unchanged

[1 mark]

Question 3

A stone of mass m is dropped from a tall building. There is significant air resistance. The acceleration of free fall is g .

When the stone reaches its terminal velocity, which information is correct?

	Magnitude of the acceleration of the stone	Magnitude of the force of gravity on the stone	Magnitude of the force of air resistance on the stone
A	zero	mg	mg
B	g	mg	mg
C	g	zero	zero
D	zero	zero	zero

[1 mark]

Question 4

A ball falls vertically and bounces on the ground

The following statements are about the forces acting while the ball is in contact with the ground

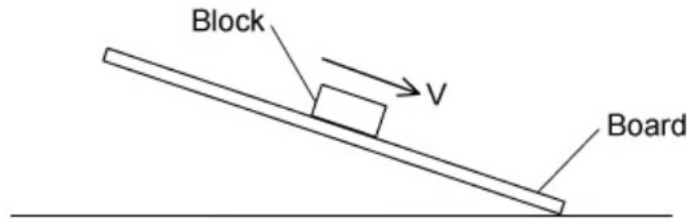
Which statement is correct?

- A. the force that the ball exerts on the ground is always equal to the weight of the ball
- B. the force that the ball exerts on the ground is always less than the weight of the ball
- C. the force that the ball exerts on the ground is always equal in magnitude and opposite in direction to the force the ground exerts on the ball
- D. the weight of the ball is always equal in magnitude and opposite in direction to the force that the ground exerts on the ball

[1 mark]

Question 5

A wooden block rests on a rough board. The end of the board is then raised until the block slides down the plane of the board at constant velocity v



Which row describes the forces acting on the block when sliding with constant velocity?

	Frictional force on block	Resultant force on block
A	down the plane	zero
B	down the plane	down the plane
C	up the plane	zero
D	up the plane	down the plane

[1 mark]

Question 6

The acceleration of free fall on planet X is $\frac{1}{6}$ of the acceleration of free fall on Earth.

The mass of a body on planet X is 60 kg.

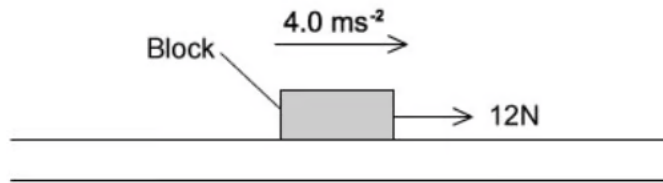
What is its weight on planet X?

- A. 10 N
- B. 59 N
- C. 98 N
- D. 360 N

[1 mark]

Question 7

A block of mass 1.20 kg is on a rough horizontal surface. A force of 12 N is applied to the block and it accelerates at 4.0 m s^{-2}



What is the magnitude of the frictional force acting on the block?

- A. 2.4 N
- B. 3.6 N
- C. 7.2 N
- D. 16.8 N

[1 mark]

Question 8

Newton's third law of motion is often summarised as

'Every action has an equal and opposite reaction.'

A cup of tea rests on a table. If the weight of the cup is the 'action' force, what is the 'reaction' force?

- A. the push of the table on the cup
- B. the push of the cup on the table
- C. the pull of the Earth on the cup
- D. the pull of the cup on the Earth

[1 mark]

Question 9

The gravitational field strength on the surface of planet R is one eighth of that on the surface of planet S.

On the surface of R, a body has its mass measured to be 2.5 kg and its weight measured to be 12.5 N

What results are obtained for measurements of the mass and weight of the same body on the surface of planet S?

	Mass on S	Weight on S
A	2.5 kg	20 N
B	20 kg	100 N
C	2.5 kg	100 N
D	20 kg	160 N

A.

[1 mark]

Question 10

Water is pumped through a hose-pipe at a rate of 80 kg per minute. It emerges from the hose-pipe horizontally with a speed of 30 m s^{-1}

Which force is required from a person holding the hose-pipe to prevent it moving backwards?

- A. 3 N
- B. 40 N
- C. 160 N
- D. 2400 N

[1 mark]

Model Answers: Medium

1

The correct answer is **B** because:

- Let's take the positive direction as the direction of the driving force and the negative direction as the direction of the resistive force
- The driving force is given as $2.0 \text{ kN} = 2000 \text{ N}$
- The resistive force is $\frac{2000}{4} = 500 \text{ N}$
- The resultant force is $2000 - 500 = 1500 \text{ N}$
- Using Newton's Second Law:
 - $F = 1500 = ma$
 - $m = \frac{F}{a} = \frac{1500}{2} = 750 \text{ kg}$

2

The correct answer is **D** because:

- If all the external forces cancel out, that means the resultant force on the body is zero
- If the resultant force is zero then the acceleration is also zero
 - $F = ma$
- Zero acceleration means that the velocity is not changing
 - $a = \frac{v}{t}$
- Hence, the momentum of the body remains unchanged
 - $p = mv$

A is incorrect because if a body is not moving then its velocity is zero, however, if acceleration is zero then that does not necessarily mean the velocity is zero. The body may be moving at a constant velocity, its acceleration would still be zero

B is incorrect because force and energy are not equivalent, so if the resultant force is zero this doesn't mean the energy will be unchanged

C is incorrect because speed is a scalar quantity, so the speed of a body may be constant but if its direction is changing then the acceleration would not be zero. Acceleration is defined as the rate of change of velocity, **not** speed

3

The correct answer is **A** because:

- At terminal velocity, the speed is constant
- So, there is no acceleration meaning the resultant force on the stone is zero
- The weight of the stone (mg) remains unchanged throughout the duration of the fall
- So, for the resultant force to be zero, the magnitude of the air resistance should be the same as the weight (mg)

4

The correct answer is **C** :

- Because Newton's Third Law states: when two bodies are in contact, the force they exert on each other is equal and opposite to each other

A is incorrect because before bouncing upwards, the ball comes to a stop and momentarily the force on the ball is reduced to zero

B is incorrect because as the ball hits the ground, the force it exerts on the ground at that moment is equal to the weight of the ball

D is incorrect because the weight of the ball is constant and always acts towards the ground, whereas the magnitude of the force on the ball changes

5

The correct answer is **C** because:

- The question tells us the block slides down the plane with constant velocity meaning there is no acceleration and hence the resultant force is zero
- Frictional forces oppose motion, and since the block moves down the plane, the friction force must act up the plane

6

The correct answer is **C** because:

- Mass is a constant quantity which does not change with a change of position or location, so the mass is 60 kg on planet X and on Earth
- The weight, however, changes depending on the strength of the gravitational field of a planet. This can be found using
 - $W = mg$
- Where $g = 9.81 \text{ m s}^{-2}$ on Earth and $g = \frac{9.81}{6} (= 1.63 \text{ m s}^{-2})$ on planet X
- The weight of the body on planet X is
 - $W = \frac{60 \times 9.81}{6} = 98 \text{ N}$

7

The correct answer is **C** because:

- Let's call the frictional force F_f
- Then, we can use $F = ma$ to equate the resultant force (= forward force – frictional force) with the mass and acceleration of the block
 - $F = (12 - F_f) = 1.2 \times 4 = 4.8$
 - $12 - 4.8 = F_f$
 - $F_f = 7.2 \text{ N}$

8

The correct answer is **D** because:

- Newton's Third Law says: when body A exerts a force on body B, body B exerts an equal and opposite force on body A of the **same type**
- For gravity, the Earth attracts the cup and the cup attracts the Earth with the same magnitude but opposite direction. Both forces are of the same type – gravitational attraction
- Consider a falling object – it is not just the object that is falling towards the earth but rather the earth too is falling towards the object. Given the huge difference in

mass, the Earth's movement towards the object is so negligible it can be ignored

A & B are incorrect because the push of the table on the cup (and vice versa) is a result of the repulsive forces between the electrons on the surface of the table and the electrons on the surface of the bottom of the cup, both of these are electrostatic forces

C is incorrect because the pull of the Earth on the cup is the same as the weight

9

The correct answer is **C** because:

- The mass of a body is constant and does not change, so it will be 2.5 kg on both planets
- The weight, however, changes depending on the strength of the gravitational field of a planet. This can be found using
 - $W = mg$
- Where g on planet S is 8 times greater than on planet R
- So, the weight of the body on planet S is
 - $W = 12.5 \times 8 = 100 \text{ N}$

A is incorrect because the mass is correct but the weight has been calculated using $2.5 \times 8 = 20$ which is wrong

B & D are incorrect because the mass should be unchanged

10

The correct answer is **B** because:

- The water that emerges from the hose has momentum, so it exerts a force on the person as it shoots out
- From Newton's Third Law, the person should exert a force of the same magnitude and in the opposite direction to prevent the hose from moving backwards

of a planet. This can be found using

- $W = mg$
- Where g on planet S is 8 times greater than on planet R
- So, the weight of the body on planet S is
 - $W = 12.5 \times 8 = 100 \text{ N}$

A is incorrect because the mass is correct but the weight has been calculated using $2.5 \times 8 = 20$ which is wrong

B & D are incorrect because the mass should be unchanged

10

The correct answer is **B** because:

- The water that emerges from the hose has momentum, so it exerts a force on the person as it shoots out
- From Newton's Third Law, the person should exert a force of the same magnitude and in the opposite direction to prevent the hose from moving backwards
- Using Newton's Second Law
 - $F = ma = \frac{m \Delta v}{t} = \frac{\Delta p}{t}$ (force = rate of change of momentum)
 - $\frac{\Delta m}{t}$ = rate of water being pumped = 80 kg / min
- In one second, $\frac{80}{60} = 1.33 \text{ kg}$
- So, 1.3 kg of water is pumped out every second
 - $\frac{\Delta m}{t}$ = rate of water being pumped = 1.33 kg s^{-1}
 - $F = \frac{\Delta p}{t} = \frac{\Delta m}{t} \times v = 1.33 \times 30 = 40 \text{ N}$

